SWYNEX INTERNSHIP

Customer Personality Analysis

Tasks 1, 2 & 3

Data Cleaning • Exploratory Data Analysis • Interactive Dashboard

Project workflow
Raw data → Cleaning → SQL analysis → Excel charts and mini-report → Interactive dashboard

A technical project report documenting the workflow, analytical approach, key findings, challenges, and recommendations.

1. Project Overview

This project was completed as part of my **Swynex Internship** and focused on analyzing customer personality and purchasing behavior using the Customer Personality Analysis dataset.

The goal was to transform raw customer data into meaningful insights that could help a business understand its customers, identify spending patterns, evaluate marketing campaign performance, and make more informed decisions.

1. **Task 1 – Data Cleaning:** Prepared the dataset using Excel Power Query.
2. **Task 2 – Exploratory Data Analysis (EDA):** Used SQL Server to explore the data and calculate metrics, then used Excel charts to visualize findings and prepare a mini-report.
3. **Task 3 – Interactive Dashboard:** Built an interactive Excel dashboard to present the analysis in a clear and accessible way.

The workflow covered the main stages of a data analysis project, from preparing raw data to communicating findings through visualizations.

2. Project Objectives

- Clean and prepare the customer dataset for analysis.
- Understand customer demographics, income, and spending behavior.
- Identify the most and least popular product categories.
- Analyze purchasing behavior across different sales channels.
- Evaluate customer responses to marketing campaigns.
- Compare spending patterns across different customer income groups.
- Create charts to communicate the findings.
- Build an interactive dashboard to support data exploration and decision-making.
- Develop practical skills in Excel, SQL Server, data visualization, and analytical thinking.

3. Dataset

Dataset: Customer Personality Analysis

Source: Customer Personality Analysis – Kaggle

The dataset contains customer-level information, including demographic characteristics, product spending, purchasing channels, and marketing campaign responses.

- **Customer demographics:** Income, education, marital status, age-related information, and other customer attributes.
- **Product spending:** Wines, fruits, meat products, fish products, sweet products, and gold products.
- **Purchasing behavior:** Purchases through the website, catalog, and physical stores, as well as purchases made using deals.
- **Marketing campaigns:** Customer responses to five marketing campaigns.
- **Customer records:** Customer identifiers and related information.

The dataset was analyzed at the customer level. Purchase counts and deal-purchase counts were interpreted according to their actual column meanings rather than assuming every record represented an individual order.

4. Tools and Technologies

Tool	Purpose
Microsoft Excel Power Query	Data cleaning and preparation
SQL Server Management Studio (SSMS)	SQL queries, calculations, and exploratory analysis
SQL	Filtering, aggregation, grouping, and comparing customer data
Microsoft Excel	Charts, visual analysis, mini-report, and interactive dashboard
GitHub	Versioned project files and technical documentation

5. Task 1 – Data Cleaning and Preparation

Objective

The first step was to prepare the raw dataset before beginning the analysis. Clean and consistently formatted data helps reduce errors and makes later calculations more reliable.

Process

I used **Excel Power Query** to inspect and prepare the dataset. The preparation stage involved reviewing data quality, checking missing values and duplicates, and ensuring the data was suitable for analysis.

1. Inspecting the dataset and understanding its columns.
2. Reviewing missing or incomplete values.
3. Checking for duplicate records.
4. Reviewing column names and data types.
5. Preparing a consistent dataset for analysis.
6. Loading the cleaned data for use in the following stages.

Challenge: Raw data may contain missing values, duplicate records, or inconsistent data formats that can affect calculations and visualizations.

Approach: I used Power Query to inspect and prepare the data before moving to SQL analysis. This separated the data preparation stage from the analytical stage and made the workflow easier to manage.

Outcome: The cleaned dataset became the starting point for the SQL exploration, Excel visualizations, and dashboard.

6. Task 2 – Exploratory Data Analysis (EDA)

After cleaning the data, I used **SQL Server Management Studio (SSMS)** to explore customer behavior and calculate key metrics. I used SQL to summarize the data, compare customer groups, and investigate patterns across products, channels, and campaigns.

6.1 Overall Business Performance

I examined customer count, total customer income, total spending across product categories, average customer spending, purchasing activity across channels, and purchases made using deals. These metrics provided an overview of the customer base and its purchasing behavior.

Important distinction: Deal purchases represent purchases associated with deals; they should not automatically be treated as total orders.

6.2 Customer Demographics and Spending

I explored how spending differed across demographic groups, including marital status and education.

- The Married group contributed the highest spending among the marital-status groups analyzed.
- The Together group was also among the leading contributors.
- The education groups showed different spending contributions, with one of the leading groups contributing more than 600,000 in spending in the analysis.

These findings describe observed spending patterns; additional analysis would be needed to determine why the differences exist.

6.3 Product Spending Analysis

The dataset contained separate spending columns for wines, fruits, meat products, fish products, sweet products, and gold products. I analyzed these columns individually and compared their totals.

- Wines had the highest total spending, exceeding 600,000 in the analysis.
- Meat products ranked second.
- Fruit products had the lowest total spending, at approximately 60,000.
- The differences between categories highlighted where customer spending was concentrated.

6.4 Purchasing Channel Analysis

I compared purchasing activity across the website, catalog, and physical stores.

- Physical stores recorded the highest purchasing activity, at approximately 12,000.
- Catalog purchases were lower, at approximately 9,000.
- Website purchases were lower still, at approximately 5,000.

These figures are approximate and should be checked against the final workbook. The results indicate the strongest channel by the purchase measure analyzed, not necessarily the channel generating the most revenue or profit.

6.5 Marketing Campaign Analysis

The dataset included response information for five marketing campaigns. I distinguished between **Accept** and **Decline** outcomes.

- Declines substantially outnumbered acceptances.
- Each campaign had many more declines than acceptances.
- The response pattern suggested that campaign effectiveness could be investigated further.

This suggested an opportunity to review campaign targeting, offers, and customer segmentation. Raw response counts are not the same as acceptance rates; percentages require comparison with the relevant number of eligible or contacted customers.

6.6 Income Groups and Average Spending

I grouped customers into three income bands: **Low:** below 30,000; **Medium:** 30,000 to 60,000; and **High:** above 60,000. I then compared the number of customers in each group with their average spending.

- The low-income group represented approximately 90% of the customer records in the analysis.
- The medium-income group showed the highest average spending among the groups analyzed.
- Customer group size alone did not explain spending behavior.

This demonstrated the value of comparing both the number of customers and their average spending instead of relying on customer counts alone.

7. Returning to Excel – Charts and Mini-Report

SQL was useful for calculating metrics and exploring relationships in the data. Excel helped communicate the results visually. I returned to Excel to create charts and organize findings into a more understandable format.

1. Overall customer and spending metrics.

2. Spending patterns across demographic groups.
3. Spending differences between product categories.
4. Purchasing behavior across sales channels.
5. Customer responses to marketing campaigns.
6. Spending patterns across income groups.

This stage connected the technical SQL analysis with business interpretation and provided a foundation for selecting the metrics and findings to highlight in the dashboard.

8. Task 3 – Interactive Dashboard

Objective

The final stage was to turn the analysis into an interactive dashboard presenting important metrics and customer behavior visually.

Dashboard development process

1. Reviewed the analysis and mini-report to identify useful findings.
2. Organized the relevant metrics and charts for presentation.
3. Built the dashboard in Excel.
4. Worked on presenting the results in a clear and accessible layout.
5. Completed the interactive dashboard as the final stage of the task.

Purpose

The dashboard brings the analysis together in one place, making it easier to explore the results without reviewing every SQL query individually. Data is prepared first, analysis produces findings, and the dashboard communicates those findings visually.

The interactive dashboard was completed as part of **Task 3 of the Swynex Internship**. A recorded walkthrough was also prepared to demonstrate the finished dashboard.

Note: Exact dashboard visuals, KPI values, and interactive controls should be listed according to the final workbook rather than assumed from the earlier EDA.

9. Challenges and How I Approached Them

Challenge	Approach
Preparing raw data	Used Excel Power Query to inspect and clean the dataset.
Understanding dataset structure	Identified columns for demographics, product spending, channels, and campaign res
Spending stored in separate columns	Calculated totals for each product category individually.
Interpreting purchase measures	Distinguished deal purchases and channel counts from total orders or revenue.
Comparing customer groups	Examined customer counts alongside average spending.
Campaign performance	Compared accept/decline responses and identified the need to evaluate acceptance
Communicating SQL results	Returned to Excel to create charts and a mini-report.
Presenting analysis clearly	Built an interactive dashboard after data preparation and EDA.

This process showed that data analysis is not only about writing queries or creating charts. It also requires understanding the dataset, choosing appropriate measures, interpreting results carefully, and communicating findings clearly.

10. Key Insights

1. **Product spending:** Wines contributed the most spending, followed by meat products, while fruits contributed the least.
2. **Purchasing channels:** Physical stores were strongest by recorded purchase activity; catalog and website activity were lower.
3. **Demographics:** Married and Together customers were among the leading contributors to total spending.
4. **Income groups:** The low-income group represented most customer records, while the medium-income group showed the highest average spending.
5. **Campaigns:** Declines were much more common than acceptances, suggesting a need to investigate targeting, offers, and engagement.
6. **Customer value:** Comparing group size with average spending revealed patterns that customer counts alone would not show.

These findings describe patterns in the dataset. They do not establish causes of customer behavior or guarantee future business results.

11. Business Recommendations

Product strategy

- Investigate why wine spending is substantially higher than other categories.
- Explore whether lower-spending categories need different promotions or positioning.
- Evaluate profitability before making decisions based on spending alone.

Sales channels

- Investigate why physical stores recorded the highest purchasing activity.
- Explore opportunities to improve website engagement and catalog performance.
- Compare channel revenue, costs, and conversion rates before allocating additional resources.

Customer segmentation

- Use income and spending patterns to identify potentially valuable customer segments.
- Avoid relying on income or demographics alone when designing offers.
- Compare average spending, total spending, and customer counts together.

Marketing campaigns

- Investigate the low number of acceptances relative to declines.
- Compare acceptance rates across campaigns and customer segments.
- Review targeting, messaging, and offers.
- Measure campaign performance using response rates and business outcomes rather than raw counts alone.

These recommendations are starting points for further investigation. Additional data and testing would be needed before implementing major business changes.

12. Final Report and Conclusion

This project followed a structured data analysis workflow, beginning with data cleaning and ending with an interactive dashboard.

First, I used Excel Power Query to prepare the customer dataset. Next, I used SQL Server Management Studio to explore customer demographics, income, product spending, purchasing channels, and marketing campaign responses.

After completing the SQL analysis, I returned to Excel to create charts and summarize the main findings in a mini-report. Finally, I used the analysis as the foundation for building the interactive dashboard.

The analysis highlighted differences in product spending, purchasing activity, demographic spending patterns, income-group behavior, and campaign responses. These findings provided potential areas for further business investigation.

The project strengthened my practical experience with data cleaning, SQL-based exploration, Excel visualization, and analytical reporting. It also reinforced the importance of interpreting results and communicating findings in a useful way.

The final outcome was a complete workflow from raw customer data to exploratory insights and an interactive dashboard.

13. Skills Practiced

- Data cleaning and preparation
- Excel Power Query
- SQL Server Management Studio
- SQL aggregation and grouping
- Exploratory data analysis
- Customer segmentation
- Data interpretation
- Excel charts and data visualization
- Dashboard development
- Business insight generation
- Technical documentation
- Communicating analytical findings

14. Suggested Project Files

The GitHub repository should contain the files needed to understand and review the work. Adjust file names and folders to match the actual repository.

```
customer-personality-analysis/
■■■ README.md
■■■ data/
■   ■■■ cleaned_customer_data.xlsx
■■■ sql/
■   ■■■ customer_analysis_queries.sql
■■■ excel/
■   ■■■ customer_analysis.xlsx
■   ■■■ interactive_dashboard.xlsx
■■■ documentation/
    ■■■ data_cleaning_notes.md
```

If the dashboard is part of the same Excel workbook as the charts, there is no need to include a duplicate workbook.

15. Dataset Attribution

The analysis uses the Customer Personality Analysis dataset available on Kaggle:

https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis

The dataset is used for educational and analytical practice as part of my Swynex Internship. Review the dataset terms and license before redistributing the original or cleaned data.

16. Project Status

- Task 1: Data cleaning and preparation — Completed
- Task 2: SQL exploratory analysis — Completed
- Task 2: Excel charts and mini-report — Completed
- Task 3: Interactive dashboard — Completed
- Dashboard walkthrough recording — Completed

Status: Completed

Thank you to Swynex for the opportunity to practice the data analysis workflow and continue developing my technical skills.